RSTGCN: Railway-centric Spatio-Temporal Graph Convolutional Network for Train Delay Prediction

Koyena Chowdhury, Paramita Koley, Abhijnan Chakraborty, and Saptarshi Ghosh

***Abstract*—Accurate prediction of train delays is critical for efficient railway operations.While earlier approaches have largely focused on forecasting the exact delays of individual trains, studies on station-level delay prediction is somewhat sparse. To address this gap, we propose the Railway-centric Spatio-Temporal Graph Convolutional Network (RSTGCN), designed to forecast average arrival delays of all the incoming trains at a particular stations for a particular time period. Our approach incorporates several architectural innovations and novel feature integrations, including train frequency–aware spatial attention, which significantly enhances predictive performance. To support this effort, we curate and release a comprehensive dataset for the entire Indian Railway Network (IRN), spanning 4,735 stations across 17 zones — the largest and most diverse railway network studied to date. We conduct extensive experiments using multiple state-of-the-art baselines, demonstrating consistent improvements across standard metrics. Specifically, RSTGCN outperforms the best baseline by 18% in MAE, 14% in MAPE, and 1–8% in RMSE on the IRN.**

***Index Terms*—Indian Railways, Train Delay Prediction, Spatio-Temporal Dependence, GNN.**

I. Introduction

Indian Railways is one of the largest railway systems globally, spanning approximately 69,000 kilometers and serving as the backbone of long-distance transportation in India. It facilitates the daily movement of more than 23 million passengers across India. Indian Railways plays a pivotal socio-economic role, enabling access to employment, education, healthcare, and markets for a significant portion of the population.

However, the vast scale and operational complexity of this mass transit system contribute to one of its most persistent challenges, *i.e., unpredictable train delays*. Although following schedules is essential for maintaining the reliability of the railway network, various operational challenges, such as network congestion, adverse weather, infrastructure constraints, and the coexistence of multiple types of trains sharing the same tracks, unlike in most other countries, often lead to significant delays. Also, several external factors, such as disruptions caused by large passenger movements during festivals, examinations and office-hours, chain pulling incidents, social or political disruptions, trespassing, and accidents may cause huge delays. In addition, the frequent introduction of new trains, often scheduled based on increasing passenger demand rather than timetable optimization, can further exacerbate congestion and cumulative delay. In India, generally punctuality yardstick is considered as 15 minutes for express trains. However, our analysis of Indian Railways data for the month of September 2024 underscores that the delays are often much more severe, revealing an average arrival delay of approximately 51 minutes, with numerous instances of delays exceeding 12 to even 24 hours. Among the delayed timestamps (*i.e.*, with delays of at least 1 minute), 35% delays are over 30 minutes, 20% over 1 hour, 10% over 2 hours, and 5% over 3.5 hours.

Disruptions in train timing are often interconnected: a single out-of-schedule train can trigger a ripple effect, delaying other trains and congesting adjacent stations. In such a heavily utilized, high-traffic, and interdependent system, accurate delay prediction can enable real-time risk assessments, dynamically adjust schedules, and mitigate delay propagation. In addition, it offers meaningful insight into the operational behavior of the rail network, providing a foundation for long-term strategic planning and improvements.

While multiple prior works have looked at predicting train-specific delays [1]–[4], *forecasting station-specific aggregated delay* can also provide great operational value, particularly in supporting station-centric tasks and policies, such as train dispatch and control decisions [5]–[7]. Hence, in this work, we address the challenge of *forecasting the average arrival delay at each station over a short-term future horizon*. Notably, prior station-based delay prediction studies have primarily been conducted in the context of the high-speed Chinese railway network [5], [6], where generally higher levels of punctuality lead to a modeling emphasis on the count of delayed trains per hour. In contrast, the Indian Railway context presents a different operational dynamics, where predicting the magnitude of delays is more relevant than simply identifying delayed instances. Accordingly, we shift the focus to modeling average arrival delays per hour. A similar formulation has also been explored in the context of the British railway network [7].

Our contributions: To address the task of predicting arrival delays, we make several important contributions in this paper: (1) We propose a Graph Convolutional Network (GCN) based spatio-temporal framework for station-wise hourly delay prediction, which we name ***R**ailway-centric **S**patio-Temporal **G**raph **C**onvolutional **N**etwork* (RSTGCN). We incorporate several domain-informed features and key architectural enhancements in our proposed model. In particular, we introduce hourly headway, a novel traffic-based feature, along with average and total arrival/departure delays. We also redesign the spatial

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attention module to integrate the train frequencies between stations and adjust the output activation for modeling delay values rather than delay counts. When evaluated on real-time large-scale temporal data from Indian Railways, our model improves over several state-of-the-art baselines.
(2) We present the first nation-wide train operational dataset for Indian Railways comprising $3,892$ long-distance trains and $4,735$ stations. This dataset is orders of magnitude larger than most existing railway traffic datasets – for instance, compared to 727 stations in [5] and 9 stations in [1], our dataset has $4,735$ stations. We will make the data available for research purposes upon acceptance of the paper.
(3) We would also like to highlight that while prior works on delay prediction in the Indian Rail network have focused on only a single route [8] or a partial region [9], to the best of our knowledge, this is the first work to study delay prediction over the entire IRN, as well as over individual zones.

II. Related Work

Train delay prediction: Existing literature on train delay prediction can be broadly classified into three categories, namely mathematical model-driven methods, statistical methods, and machine learning methods [10]. *Mathematical model-driven methods* [11]–[13] primarily attempt to simulate the train operations and delay propagation; however, these models rely heavily on the underlying mathematical assumptions and fail to exploit the train delay data, leading to limited generalization performance [14], [15]. *Statistical methods* are primarily developed around Bayesian network based framework, exploring the effect of prediction horizon and train information [16], complexity and dependency of train operations [17], modeling delay propagation and clustering delay patterns [18], along with other approaches as fuzzy petri-nets [19], which is based on the Markov assumption. Among recent Markov chain based works, Shahin et al. [20] explored train delay prediction with homogeneous Markov chains, while Xu et al. [21] proposed a delay prediction model based on non-homogeneous Markov chain with sparse transition matrix modeled by Gaussian kernel density estimation. Spanninger et al. [22] further employed advanced Markov chains setting based on process time deviations instead of absolute delays.
Finally, *machine learning models* employed various modeling paradigms for addressing train delay prediction, including decision tree [23], ANN [24], shallow and deep extreme learning machines [25]. Among the sequential models, Wen et al. [26] used LSTM to model delay-influencing factors on Dutch railways. Temporal-only approaches, such as Ping et al. [27], leverage RNNs, while Oneto et al. [28] employ data-driven models based solely on historical data, overlooking spatial correlations. Huang et al. [3], [4] proposed FCLL-Net and CLF-Net, combining LSTM, CNN, and FNN to capture temporal, spatio-temporal, and external operational features. Recently, Li et al. [29] proposed TLF-net to capture the effects of rolling stock connections and terminal route conflicts. Recently, Zhang et al. [30] employed a temporal point process–based framework (TANTPP), Gao et al. [31] and Chtioui et al. [32] applied XGBoost, and Artan et al. [33] utilized Markov chain models for predicting train delays.

GNN-based models: While the above works ignore the spatial dependency in the network, recent works employ tools from graph neural network literature to exploit complex spatio-temporal dependency. Among them, GCRNN [34] and STGCN [35] combine GCNs with recurrent or temporal convolution layers. ASTGCN [36] introduced attention mechanisms, while TSTGCN [5] further incorporated inter-station distances. Heglund et al. [7] applied STGCN to the British railway; whereas Li et al. [2] introduced SAGE-Het for modeling heterogeneous interactions. Zhang et al. proposed MATGCN [37], followed by C-MATGCN [6] using graph communities and fuzzy trees. MATGCN [37] combined some external factors, such as weather and holidays, using DLP and employed a multifeature attention mechanism to capture their impact on train delays. Recently, DB-STGCN [1] incorporated domain knowledge via Bayesian causality graphs. Among graph-embedding based models, Tang et al. [38] attempted to embed a train's route in a network topology perspective towards reducing feature dimension and improving accuracy. Among these prior works, [5]–[7], [37] focus on station-specific delay prediction, while the others are train-specific. Note that, we exclude [1], [6], [37] from baseline comparisons due to their reliance on weather data, which is not available for IRN.

Traffic flow prediction is similar well-studied problem in spatio-temporal modeling, whose solutions often apply to train delay prediction without major modifications. Early approaches include spatio-temporal GCNs [39] and attention mechanisms for long-term dependencies [36]. Further developments have explored dynamic spatial modeling through Laplacian estimators [40], temporal GCNs [41], adaptive graph learning [42], and the integration of networking principles [43]. Recent advances include STMGCN [44] for multi-correlation modeling, STCL-AGA [45] with graph augmentation and masking, DAGCRN [46] with adaptive adjacency and temporal attention, STGSA [47] for localized and long-term dependency capture, and STFTGCN [48], which integrates trend-aware spatio-temporal features.

Indian railway network analysis: Most of the above train delay prediction works focus on the Chinese rail network, and there has been few attempts on analyzing IRN. Among them, the notable works include statistical delay analysis for express trains [49], arrival delay prediction via a zero-shot framework in partial region with 135 trains [9], and delay prediction in the Delhi-Mumbai route [8]. However, none of these works addressed delay prediction across the entire IR network or different zones, highlighting the need for a system-wide study.

Note that, none of the above approaches takes into account the influence of train counts between stations on delay propagation. To bridge this gap, we enhance the spatial attention of the GNN and also introduce hourly headway as a novel feature in the present work.

III. Dataset description

We develop a railway network dataset based on train operation records from the Indian Railway (IR) system, including data from $3,892$ long-distance passenger trains (excluding freight trains and 'local' trains which travel

TABLE I: Examples of train running records in our dataset

Date	Train No.	Train Name	Code	Station	Dist.	Sch. Arr.	Act. Arr.	Arr. Delay	Sch. Dep.	Act. Dep.	Dep. Delay
7 Sept. 2024	12277	Satabdi Express	KGP	Kharagpur	116 KM	03:55 PM	04:19 PM	24M Late	04:00 PM	04:24 PM	24M Late
7 Sept. 2024	12277	Satabdi Express	BLS	Balasore	237 KM	05:15 PM	05:35 PM	20M Late	05:20 PM	05:40 PM	20M Late
25 Sept. 2024	12951	NdlsTejasRaj Exp.	BVI	Borivali	30 KM	05:22 PM	05:40 PM	18M Late	05:24 PM	05:42 PM	18M Late
25 Sept. 2024	12951	NdlsTejasRaj Exp.	ST	Surat	262 KM	07:43 PM	07:53 PM	10M Late	07:48 PM	07:58 PM	10M Late

only in the vicinity of metropolitan cities). This dataset is sourced from https://runningstatus.in/, which provides detailed information such as running date, train number, train name, station code, station name, distance from source, scheduled and actual arrival and departure times along with delays for various express trains over a specified time period, as shown in Table I. The train running information spans the period September 01-30, 2024. We chose September as a representative month as it avoids extreme climate conditions (neither peak-monsoon, nor the winter period), and is without any major festivals in India. Our dataset contains 1,282,326 train records, where each row corresponds to a train at a specific station on a given day. We exclude instances where no train is scheduled or where no delay is recorded for the corresponding time interval.

The Indian Railways Network dataset: We represent the IR system by a network (which we call the IR network or IRN) where the nodes/vertices are the stations, and there is an edge between two nodes s_i and s_j if and only if they are *consecutive* stations on at least one train route. In other words, there is a link/edge (s_i, s_j) only if there is a direct track/line connecting the two stations. The IRN thus formed comprises $4,735$ stations (the major train stations in India) connected by $9,336$ edges (direct lines). The key statistics of the IR network in our dataset are summarised in Table II.

TABLE II: Statistics of Indian Railways

Description	Value	Description	Value
No. of Stations	4735	No. of Trains	3892
No. of Edges	9336	Avg Node degree	3.94
Avg Trains per Station	17.95	Avg Station Halts per Train	21.84
Avg. distance (KM)	38.05	Avg. trains per edge	8.69
Avg Headway (mins)	48.76	Median Headway (mins)	14.0

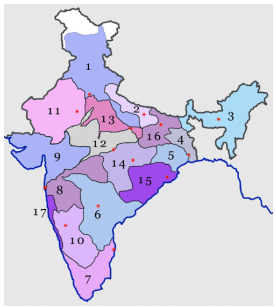


Fig. 1: Zones in the Indian Railways

Zone (Code) - Number of Stations
1. Northern Railway (NR) - 449
2. North Eastern Railway (NER) - 218
3. Northeast Frontier Railway (NFR) - 286
4. Eastern Railway (ER) - 236
5. South Eastern Railway (SER) - 205
6. South Central Railway (SCR-499)
7. Southern Railway (SR) - 316
8. Central Railway (CR-) 272
9. Western Railway (WR) - 424
10. South Western Railway (SWR) - 237
11. North Western Railway (NWR) - 334
12. West Central Railway (WCR) - 216
13. North Central Railway (NCR) - 240
14. South East Central Railway (SECR) - 113
15. East Coast Railway (ECOR) - 264
16. East Central Railway (ECR) - 354
17. Konkan Railway (KRCL) - 72

Zones in the IRN: For organizational convenience at different levels, the IR network is divided into 17 major regions called 'zones' [50]. This zonal division allows the extremely large system to balance technical, geographic, financial, and organizational factors, making the national railway more scalable, coordinated, and responsive to regional needs. The zones vary in size and number of stations, ranging from South Central Railway (SCR) with 499 stations to Konkan Railway Corporation Limited with only 72 stations. Fig 1 gives a map of the IRN showing the zones, and also states the name and number of stations of each zone. Our model is evaluated both on the full network and separately for each zone, to ensure robust validation across diverse network scales.

IV. Problem Statement

A railway network is defined as a undirected graph $G = (S, E, A, M, K)$, where S denotes the set of stations with $|S| = N$, E denotes set of edges, where each edge denotes existence of a track between two consecutive stations. A is the adjacency matrix which denotes connectivity between stations. M denotes the distance matrix between stations. K is the train frequency matrix, where k_{ij} denotes the number of distinct trains between the adjacent stations i and j. Each train r has a schedule that specifies exact arrival time t_r^a and departure time t_r^d for each station. Suppose, the train actually reaches the station at $\bar{t}_r^a$ and departs at $\bar{t}_r^d$. Then arrival delay Δ_a is defined as the difference between actual arrival time and scheduled arrival time, $\Delta_a = \bar{t}_r^a - t_r^a$. Similarly, departure delay is defined as the difference between actual departure time and scheduled departure time, $\Delta_d = \bar{t}_r^d - t_r^d$. In our work, we are particularly interested in *hourly arrival delay (Δ_a) prediction at each station averaged across the trains arriving in a particular time period (e.g., a particular hour).*

Given a fixed time period τ, let F be the number of features per station (for our method, $F = 5$). Each station records several features capturing various train running statistics during τ, which we will discuss in detail in the following section V-A. The feature matrix for all stations is denoted as $X_\tau = (X_\tau^1, X_\tau^2, \ldots, X_\tau^N) \in \mathbb{R}^{N \times F}$, where $X_\tau^i \in \mathbb{R}^F$ represents the feature vector of station i. We set $y_t^i \in \mathbb{R}$ as the average arrival delay at station i in the future time period t.

Problem: Given a railway network G, the objective is to predict the average arrival delay at all stations in the network for the next t_p time steps. $Y = (y_1, y_2, \ldots, y_N)^t \in \mathbb{R}^{N \times t_p}$, based on the historical and node-specific data of all stations S over a past time period τ. Here, $y_i = (y_i^{\tau+1}, y_i^{\tau+2}, \ldots, y_i^{\tau+t_p})$ represents the predicted average delay sequence at station i over the future time window of length t_p.

V. RSTGCN: Proposed Framework

In this section, we propose RSTGCN (Railway-centric Spatio-Temporal Graph Convolutional Network) for predicting the node-wise average delay in a rail network. Our model borrows some components from TSTGCN [5], but modifies

the components/features to make them more suitable for the Indian Railway network. We start by describing the station-specific features that we propose to incorporate in our model.

A. Station-specific features

As our primary task is to predict station-specific average arrival delay, we propose to maintain the following two features per station, namely, (1) **hourly average arrival delay**, and (2) **hourly average departure delay**. Prior works employed two different station-level features – the number of arrival and departure delays of trains per hour [5]. As our prediction problem has shifted from predicting the number of delayed trains per station to predicting the average delay prediction per station, we have modified these features as stated above. We further argue that not only the average delays, but **total hourly arrival and departure delays** at each station also play important role in the delay modeling. So, for each station, we also maintain total hourly arrival and departure delays.

Moreover, we propose to employ *a novel station-specific feature*, called **hourly headway**. Hourly headway is defined as the average difference between consecutive trains according to the actual schedules for a given hour at a specific station. Prior studies have shown that train delays can be affected by headway [51]. Our hypothesis is that short headways, indicating more frequent trains, hinder the recovery of delays for subsequent incoming trains. In contrast, longer headways provide greater buffer time, enabling better delay mitigation.

B. Model details

Our framework is composed of three distinct modules to process the recent, daily, and weekly history, respectively. Each module consists of multiple spatio-temporal encoder blocks, where each block consists of an attention block, followed by a convolution block, and finally concluded by a fully connected layer, as shown in Fig 2. The outputs of these three modules are linearly combined to obtain the final output. Now, we describe the framework in detail, highlighting our model-specific contributions.

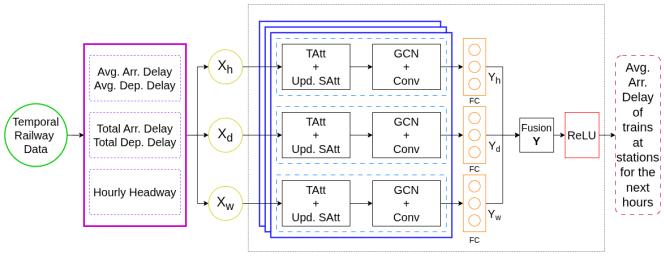


Fig. 2: RSTGCN model architechture

Processing input data: The training data consists of the average arrival delay at the nodes of the network at different timestamps, which is a common time-series data in a graph network. With sampling frequency q per day and prediction window t_p, we extract past graph signals X_τ over t_h, t_d, and t_w as inputs to the recent, daily, and weekly components. The temporal timestamp inputs are described as follows:

(1) Recent history: The arrival delay of a train at one station increases the chance of higher arrival delay of upcoming trains at that station. To encode this, for each node, we take as input the recent history, formed by the average arrival delay at previous τ hours at that node. From the network perspective, the recent history is expressed as:

$$\mathbf{X}_h = (X_{t_0-t_h+1}, X_{t_0-t_h+2}, \ldots, X_{t_0}) \in \mathbb{R}^{N\times F\times t_h}$$

(2) Daily history: Delay varies based on the time of the day. Many trains run on a daily basis, following the same time schedule everyday. Hence, there is a similar rail traffic at a particular time of the day, which plays a significant role in deciding the arrival delay at that hour. For instance, Fig. 3a shows the variation of mean arrival delays of trains on two consecutive days at Howrah (HWH) Junction, a major station in IRN. The two days exhibit similar patterns at many time instances, indicating strong temporal dependence in delay patterns (though there are variations at certain instances as well). Therefore, the daily average delay at the particular hour t is passed as input:

$$\mathbf{X}_d = (X_{t_0-(t_d/t_p)\cdot q+1}, \ldots, X_{t_0-((t_d/t_p)-1)\cdot q+1}, \ldots, X_{t_0-q+1}, X_{t_0-q+t_p}) \in \mathbb{R}^{N\times F\times t_d}$$

(3) Weekly history: In India, many express trains run on a weekly basis. Therefore, accounting for rail traffic at a particular hour for the past few weeks is important. For instance, Fig. 3b shows the variation of mean arrival delays of trains at Howrah (HWH) Junction, on two days separated by a week (September 5 and 12, 2024). The delay peaks observed at specific hours on a given day tend to recur at the same hours on the corresponding day of the following week, demonstrating strong weekly temporal dependence. Towards that, the following weekly history is taken into account to capture weekly rail traffic characteristics:

$$\mathbf{X}_w = \big(X_{t_0-7\cdot(t_w/t_p)\cdot q+1}, \ldots, X_{t_0-7\cdot((t_w/t_p)-1)\cdot q+1}, \ldots, X_{t_0-7\cdot q+t_p}\big) \in \mathbb{R}^{N\times F\times t_w}$$

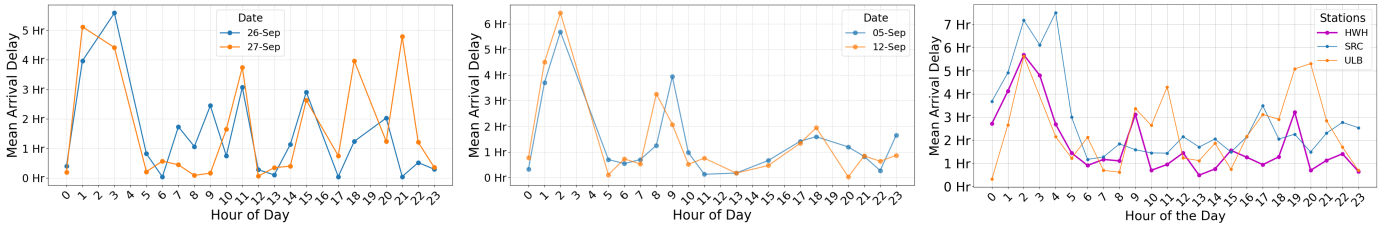
For example, to predict the next 3 hours of delay ($t_p = 3$) at a particular station on Sunday, 21st September at 8:00 AM, say, we set $t_h = 3$, $t_d = 1$, and $t_w = 1$. This means the model will use the past 3 hours of delay data from 5:00 AM to 8:00 AM on the same day, the delay data from the same hour (8:00 AM) on the previous day, *i.e.*, Saturday, 20th Sept., and the delay data from 8:00 AM on the previous Sunday, *i.e.*, 14th September, all for the same station.

Layers for further processing: The input is then passed through the following layers for further processing.

Spatio-temporal attention: This module captures long and short-term spatio-temporal dependencies. Generally, several layers of multi-head attention modules are employed here, as follows.

(1) *Temporal attention:* This module computes the effects of historical delay at a specific node at various temporal granularities. We use a time-based self-attention mechanism, where the time weight matrix Z captures dependencies between time steps i and j.

$$Z = V_t \cdot \sigma\left(((XU_1)U_2) \odot (U_3X) + b_t\right)$$



(a) (b) (c)

Fig. 3: **(a) Daily (temporal) correlations:** Mean arrival delay (hourly) over two consecutive days at Howrah (HWH) Junction, **(b) Weekly (temporal) correlations:** Mean arrival delay (hourly) across two Thursdays in consecutive weeks at HWH Junction. (c) **Spatial correlation:** Mean arrival delay (hourly) of HWH and some neighbouring stations. High delays at HWH often correspond to high delays at some adjacent station. While Figure 3a and 3b justify the usage of temporal features (daily and weekly history), Figure 3c advocates for the use of spatial attention.

where $X = (X_1, X_2, \ldots, X_{T_{r-1}})$ is the input to the r-th spatio-temporal module. F_{r-1} and T_{r-1} denote the number of features and time steps at input layer r, respectively. V_t and b_t are learnable projection parameters. Finally we get Z' by applying $Softmax$. This temporal attention matrix Z' is next forwarded to the input of the r-th layer of the spatial attention module.

(2) *Spatial attention:* Delay at a given node can propagate to neighboring nodes in subsequent time steps. For instance, Fig. 3c shows the variation of mean arrival delays at Howrah junction (HWH) and a few neighbouring stations. The figure illustrates that high delays (spikes) at HWH often coincide with corresponding spikes at neighboring stations, indicating strong spatial correlations across adjacent nodes. To effectively capture this spatial dependency, we employ a spatial attention module. Our updated formulation considers both the distance and the number of connecting trains between stations as key factors influencing delay prediction. Here, a linear transformation is applied to the input features and the correlation matrix C:

$$C = V_S \cdot \sigma\left(((X_{Z'} \cdot W_1)W_2) \odot (W_3 X_{Z'}) + b_S\right)$$

where $X_{Z'}$ denotes the input data from the time attention module, and W_1, W_2, and W_3 are feature transformation matrices. The distance weight matrix $M \in \mathbb{R}^{N \times N}$ assigns $M_{ij} = \frac{1}{d_{S_i S_j}}$ for neighbors and 0 otherwise, giving higher weight to closer stations and reducing delay influence over longer distances [5]. We propose that the number of trains passing between a pair of stations also plays a crucial role in deciding the degree of delay propagation. Specifically, if there are too many trains between two stations, the recovery opportunities for delays are limited. On the contrary, fewer trains between stations would have a higher chance of compensating for the delay. To capture this intuition, we modify M as

$$M_{ij} = \frac{1}{d_{S_i S_j}} \times \frac{k_{S_i S_j}}{k_{max}} \tag{1}$$

where k_{max} is the maximum number of trains between any station-pair in that network. The denominator term is used to normalize the multiplicative quantity. The final spatial attention matrix Q is obtained by employing $Softmax$ on element-wise product of the correlation weight matrix C and the weight matrix M, such that, $Q = Softmax(C \odot M)$.

Graph convolution: Node encodings from spatial attention module is further fed into the graph convolution network. As the input delay data is network data, a graph neural network (GNN) is employed to obtain node representations. Following [5], the data is transferred to the spectral domain using Chebyshev's polynomial, where the convolution kernel is employed. The graph convolution operation generates node representations, collecting adjacent node information.

2D-CNN: After the GCN network, a two-dimentional CNN is employed to merge the information of the nodes for adjacent time slices. For example, the convolution at the r-th layer of the daily component is given by:

$$X_d^{(r)} = \mathrm{ReLU}\left(\Phi * \left(\mathrm{ReLU}\left(g_\theta *_G \hat{X}_d^{(r-1)}\right)\right)\right)$$

where Φ is the convolution kernel parameter.

Integration of all components: During fusion, each node assigns different importance to the three components, with weights learned from historical train data. The final integrated output is: $\hat{\mathcal{Y}} = W_h \odot \hat{Y}_h + W_d \odot \hat{Y}_d + W_w \odot \hat{Y}_w$ where W_h, W_d, and W_w are learnable parameters. $\hat{Y}_h$, $\hat{Y}_d$, and $\hat{Y}_w$ denote the final outputs of the recent, daily, and weekly components after the 2D-CNN.

Final Layer: To ensure non-negative outcomes to predict the average arrival delay, we propose to employ a ReLU operator in the final outcome of this layer. We modify $\hat{Y}$ as: $\hat{Y} = ReLU(\hat{\mathcal{Y}})$. In the Indian railway system, trains typically arrive either on time or with a delay, while early arrivals are extremely rare. To better reflect this operational reality and reduce noise in the data, we eliminate negative delay values at prediction time. This ensures that the model focuses solely on predicting delays, aligning with real-world patterns, and improving the robustness of the prediction task.

VI. Experimental Evaluation

We validate our model by performing several experiments over the Indian Railway network (introduced in Section III).

A. Data processing and splits

As stated earlier, our collected data spans from September 1 to 30, 2024. We preprocess the data by normalizing delays

from the unit of minutes to hours. Next, we compute the hourly average arrival delay per station over $30 \times 24 = 720$ hours. Hours with no train arrivals at a particular station are excluded from prediction but are recorded in the historical data for consistency. Extreme delay cases (≥ 24 hours) were retained to preserve the true distribution of operational disruptions.

Data splits: The chronologically first 168 timestamps (*i.e.*, the first 7 days up to 8th September 2024, 12:00 AM) are used solely as historical delay input and are excluded from both training and testing. Out of the remaining data points (spanning 23 days), we use the chronologically first 70% data for training, 10% for validation (for deciding hyperparameters), and the final 20%, starting from 26th September 2024 at 08:00 AM, is reserved for testing. A sliding window approach is used to incorporate past delay history into the model input.

B. Metrics

To assess the performance of our model, we employ the following evaluation metrics, namely, 1) Mean absolute error (MAE), 2) Mean Absolute Percentage Error (MAPE), and 3) Root Mean Square Error (RMSE). To avoid division by zero in the case of MAPE, if there is no delay, we just take the percentage differences between the predicted and actual values. All metrics are reported after averaging across all the stations in the network, across all time stamps in the test set.

C. Baselines

We compare our model with the following baselines:

1) Historical Average (HA): Computes a simple historical average over the input window.
2) Long short-term memory network (LSTM) [52] [3]: LSTMs can efficiently learn temporal dependencies by modeling the delay sequence at each station.
3) GRU [53] [52]: Similar to LSTM, but with a simplified architecture, GRUs mitigate the vanishing gradient issues in traditional RNNs.
4) Random forest (RF) [54] [55]: RF has shown strong performance in train delay prediction tasks.
5) Spatial-Temporal Graph Convolutional Network (STGCN) [7]: Captures cascading delays by modeling temporal dynamics alongside spatial correlations among stations in the railway network.
6) ASTGCN [36]: Incorporates both temporal and spatial attention mechanisms, and has been applied successfully to traffic flow prediction.
7) TSTGCN [5]: A spatio-temporal attention model specifically designed for railway delay prediction, leveraging both temporal and spatial patterns.

D. Experimental setup and Hyper-parameter tuning

Graph-based models: All Graph-based models (STGCN, ASTGCN, TSTGCN, RSTGCN) are implemented with the MXNet package (https://mxnet.apache.org/) on a system with 48 CPU cores. To ensure fairness, we adopt an identical architectural configuration for all these models, which is adopted from TSTGCN [5]. The order of Chebyshev polynomials is set to 3, and all convolutional layers utilize 64 kernels. Each temporal convolution layer also employs 64 kernels.

TABLE III: Range of hyper-parameters for non-graph baselines

LSTM	GRU	RF
Hidden units: [50, 75, 100, 150]	Units: [32, 64, 100]	N Estimators: [100, 200, 400]
LR: [0.001, 0.0005, 0.0002]	LR: [0.001, 0.0005]	Max Depth: [10, 20, None]
Batch Size: [32, 64, 128]	Batch: [64, 128]	Min Samples Split: [2, 5, 10]
	Dropout: [0, 0.2]	Min Samples Leaf: [1, 2, 4]
		Max Features: [sqrt, log2]

Non-graph methods: For non-graph baselines (LSTM, GRU, and RF), we have selected the hyper-parameters (e.g., # hidden units, batch size, and LR for LSTM) from a wide range of values. Table III presents the range of hyper-parameters. To ensure fairness, we select hyper-parameters for each zone distinctly, based on the best performance over validation data.

All models are trained to predict aggregated arrival delay for each station in the network for 1 hour, 2 hour and 3 hour interval window for every hour in the test horizon. Models are trained using mean squared error (MSE) as the loss function, optimized via backpropagation. During training, a batch size of 4 and a learning rate of 0.001 are used. Early stopping is used to stop overfitting for all models.

For temporal input segmentation, we set the lengths of the three components as $T_h = 3$, $T_d = 1$, and $T_w = 1$. The size of the prediction window is set to $T_p = 3$, which means that the model predicts the arrival delays for a particular station for the next 3 hours, using data from the previous three hours, the corresponding hour of the previous day, and the same hour of the previous week.

E. Delay prediction results

Zone-wise results: Table IV, Table V, and Table VI report MAE, MAPE, and RMSE of train delay predictions for the next 1, 2, and 3 hours, respectively for all the baselines and the proposed model RSTGCN. The zones are ordered in descending order of the number of stations (largest zones at the top).

Among the baselines, traditional temporal-only models (HA, LSTM, GRU and RF) generally perform inferior to graph-based spatio-temporal variants, particularly in short-horizon predictions. HA, the simple average-based baseline, exhibits the weakest overall performance. LSTM provides modest improvements over HA, however the improvement is limited by its limitations in capturing spatial dependencies. GRU, a more sophisticated temporal variant, achieves minor improvement over LSTM but also doesn't consider the spatial relationships. RF outperforms other traditional models and is competitive for shorter horizons, but falls short compared to graph-based deep learning approaches.

Graph-based models, namely STGCN, ASTGCN, and TSTGCN significantly outperform traditional baselines by leveraging spatio-temporal relationships. Among them, STGCN demonstrates consistent and balanced performance, but its effectiveness is limited due to the absence of an attention mechanism. ASTGCN improves upon STGCN, by providing additional flexibility in long-term dependency modeling via the attention mechanism. TSTGCN further improves upon ASTGCN, alongside achieving the best performance among

TABLE IV: Zone-wise Mean Absolute Error (MAE) across different delay prediction models. Each row corresponds to a zone, and gives the aggregated average arrival delay error (in hours) over all stations within that zone. Each column is for a prediction model. For each model, results are shown for three prediction horizons: upto 1-hour / 2-hour / 3-hour. Best values are in bold.

Zones	HA	LSTM	GRU	RF	STGCN	ASTGCN	TSTGCN	RSTGCN (ours)
SCR	0.127 / 0.110 / 0.105	0.117 / 0.109 / 0.105	0.114 / 0.100 / 0.096	0.104 / 0.096 / 0.092	0.096 / 0.099 / 0.099	0.087 / 0.094 / 0.095	0.079 / 0.088 / 0.089	**0.061 / 0.071 / 0.076**
NR	0.187 / 0.167 / 0.159	0.212 / 0.200 / 0.192	0.173 / 0.156 / 0.150	0.158 / 0.145 / 0.140	0.124 / 0.126 / 0.126	0.127 / 0.130 / 0.131	0.110 / 0.122 / 0.125	**0.098 / 0.103 / 0.104**
WR	0.066 / 0.057 / 0.054	0.095 / 0.088 / 0.083	0.076 / 0.068 / 0.065	0.055 / 0.049 / 0.047	0.041 / 0.040 / 0.040	0.043 / 0.041 / 0.039	0.036 / 0.040 / 0.039	**0.029 / 0.032 / 0.033**
ECR	0.188 / 0.166 / 0.158	0.222 / 0.210 / 0.202	0.197 / 0.173 / 0.167	0.155 / 0.142 / 0.137	0.114 / 0.113 / 0.114	0.120 / 0.128 / 0.128	0.106 / 0.123 / 0.126	**0.095 / 0.106 / 0.108**
NWR	0.079 / 0.065 / 0.062	0.080 / 0.071 / 0.068	0.070 / 0.058 / 0.056	0.061 / 0.053 / 0.050	0.041 / 0.041 / 0.041	0.043 / 0.041 / 0.039	0.041 / 0.041 / 0.040	**0.035 / 0.035 / 0.034**
SR	0.117 / 0.107 / 0.104	0.131 / 0.134 / 0.132	0.106 / 0.099 / 0.097	0.092 / 0.088 / 0.087	0.090 / 0.094 / 0.094	0.078 / 0.084 / 0.087	0.068 / 0.083 / 0.088	**0.054 / 0.063 / 0.067**
NFR	0.098 / 0.081 / 0.075	0.094 / 0.084 / 0.080	0.087 / 0.072 / 0.067	0.082 / 0.071 / 0.067	0.052 / 0.052 / 0.052	0.062 / 0.057 / 0.055	0.054 / 0.054 / 0.054	**0.042 / 0.046 / 0.046**
CR	0.193 / 0.172 / 0.164	0.248 / 0.235 / 0.223	0.170 / 0.158 / 0.156	0.165 / 0.153 / 0.149	0.140 / 0.145 / 0.146	0.142 / 0.146 / 0.147	0.109 / 0.131 / 0.137	**0.102 / 0.110 / 0.116**
ECOR	0.131 / 0.110 / 0.104	0.136 / 0.122 / 0.116	0.122 / 0.104 / 0.099	0.108 / 0.097 / 0.092	0.084 / 0.085 / 0.084	0.092 / 0.091 / 0.091	0.080 / 0.084 / 0.087	**0.065 / 0.072 / 0.073**
NCR	0.185 / 0.157 / 0.149	0.179 / 0.166 / 0.160	0.162 / 0.139 / 0.133	0.148 / 0.133 / 0.128	0.115 / 0.116 / 0.116	0.121 / 0.120 / 0.125	0.110 / 0.113 / 0.115	**0.096 / 0.104 / 0.109**
SWR	0.077 / 0.066 / 0.063	0.074 / 0.071 / 0.067	0.068 / 0.059 / 0.057	0.060 / 0.054 / 0.051	0.050 / 0.049 / 0.049	0.049 / 0.052 / 0.052	0.047 / 0.048 / 0.050	**0.034 / 0.036 / 0.039**
ER	0.109 / 0.097 / 0.093	0.143 / 0.137 / 0.134	0.109 / 0.100 / 0.098	0.087 / 0.079 / 0.076	0.056 / 0.057 / 0.057	0.071 / 0.070 / 0.069	0.057 / 0.062 / 0.063	**0.042 / 0.048 / 0.048**
NER	0.189 / 0.165 / 0.158	0.198 / 0.188 / 0.185	0.163 / 0.141 / 0.135	0.152 / 0.133 / 0.127	0.095 / 0.094 / 0.094	0.108 / 0.102 / 0.103	0.099 / 0.098 / 0.100	**0.078 / 0.079 / 0.084**
WCR	0.189 / 0.167 / 0.161	0.187 / 0.178 / 0.175	0.168 / 0.150 / 0.146	0.156 / 0.143 / 0.139	0.152 / 0.151 / 0.150	0.146 / 0.147 / 0.145	0.116 / 0.132 / 0.136	**0.098 / 0.114 / 0.118**
SER	0.172 / 0.147 / 0.138	0.184 / 0.168 / 0.163	0.158 / 0.136 / 0.129	0.147 / 0.128 / 0.121	0.095 / 0.097 / 0.097	0.116 / 0.111 / 0.111	0.106 / 0.106 / 0.108	**0.088 / 0.090 / 0.095**
SECR	0.238 / 0.207 / 0.199	0.232 / 0.214 / 0.211	0.214 / 0.189 / 0.183	0.197 / 0.179 / 0.175	0.169 / 0.167 / 0.167	0.176 / 0.165 / 0.169	0.138 / 0.135 / 0.143	**0.113 / 0.120 / 0.134**
KRCL	0.217 / 0.185 / 0.174	0.187 / 0.169 / 0.164	0.184 / 0.160 / 0.153	0.165 / 0.151 / 0.146	0.160 / 0.162 / **0.159**	0.182 / 0.186 / 0.189	0.169 / 0.184 / 0.185	**0.133 / 0.156** / 0.163

TABLE V: Zone-wise Mean Absolute Percentage Error (MAPE %) across different delay prediction models

Zones	HA	LSTM	GRU	RF	STGCN	ASTGCN	TSTGCN	RSTGCN (ours)
SCR	24.09 / 20.63 / 19.37	21.56 / 20.36 / 19.41	20.22 / 17.80 / 17.00	20.63 / 18.52 / 17.72	15.29 / 16.29 / 16.24	10.15 / 11.22 / 11.56	8.83 / 10.30 / 10.59	**7.11 / 8.51 / 9.41**
NR	41.82 / 36.17 / 34.02	55.24 / 50.05 / 47.38	40.16 / 35.05 / 33.54	36.29 / 32.58 / 31.16	25.01 / 26.29 / 26.12	17.16 / 17.89 / 18.14	13.39 / 15.89 / 16.70	**12.22 / 13.48 / 14.05**
WR	20.17 / 16.84 / 15.85	36.90 / 32.44 / 29.60	26.20 / 22.78 / 21.43	16.55 / 14.03 / 13.26	8.51 / 8.78 / 8.64	6.59 / 6.31 / 6.11	4.87 / 5.80 / 5.88	**4.67 / 5.13 / 5.41**
ECR	35.04 / 30.52 / 29.26	51.44 / 47.70 / 44.38	43.07 / 36.95 / 34.84	29.85 / 27.10 / 26.15	19.95 / 20.78 / 20.68	13.32 / 15.10 / 15.15	11.54 / 14.71 / 15.04	**10.19 / 12.50 / 12.73**
NWR	20.51 / 16.06 / 14.88	22.53 / 19.21 / 17.69	16.50 / 13.30 / 12.40	13.83 / 11.42 / 10.64	8.75 / 9.11 / 8.99	6.65 / 6.13 / 5.82	5.63 / 5.70 / 5.70	**5.47 / 5.33 / 5.22**
SR	35.96 / 31.79 / 30.34	45.69 / 46.33 / 44.29	32.25 / 28.93 / 28.14	30.16 / 27.69 / 26.93	22.22 / 23.74 / 23.86	12.43 / 14.29 / 14.59	9.57 / 13.72 / 14.50	**8.74 / 11.07 / 11.52**
NFR	17.96 / 14.31 / 13.01	17.02 / 14.53 / 13.70	12.41 / 10.43 / 10.01	14.35 / 12.02 / 11.15	7.01 / 7.03 / 7.22	7.07 / 6.57 / 6.27	5.94 / 6.04 / 6.14	**4.72 / 5.35 / 5.29**
CR	37.06 / 32.80 / 31.24	56.92 / 51.51 / 47.66	26.98 / 27.14 / 28.52	33.42 / 31.06 / 30.38	26.24 / 27.92 / 27.54	17.03 / 17.74 / 17.87	12.09 / 15.42 / 16.31	**11.46 / 12.99 / 13.81**
ECOR	25.74 / 21.72 / 20.57	30.28 / 26.10 / 24.32	24.48 / 21.53 / 20.68	22.54 / 20.43 / 19.61	15.52 / 15.45 / 15.15	11.06 / 11.08 / 11.17	9.10 / 9.79 / 10.49	**7.47 / 8.59 / 8.94**
NCR	31.08 / 26.33 / 25.07	31.87 / 29.80 / 28.28	25.32 / 22.43 / 21.62	24.77 / 22.46 / 21.56	17.43 / 17.64 / 17.40	13.12 / 13.03 / 14.04	11.48 / 11.86 / 12.52	**9.97 / 10.96 / 11.93**
SWR	20.23 / 16.52 / 15.09	19.49 / 18.05 / 16.70	14.72 / 12.63 / 12.01	13.67 / 11.77 / 11.00	8.36 / 8.85 / 8.78	6.95 / 7.72 / 7.85	6.48 / 6.94 / 7.23	**4.69 / 5.47 / 5.95**
ER	31.42 / 26.59 / 24.73	48.84 / 43.45 / 40.37	31.01 / 27.36 / 26.47	20.66 / 18.34 / 17.66	11.08 / 12.34 / 12.07	9.61 / 9.76 / 9.88	7.01 / 8.33 / 8.68	**5.86 / 6.93 / 7.18**
NER	31.89 / 26.49 / 25.05	37.16 / 33.67 / 32.13	21.98 / 19.12 / 18.17	19.92 / 17.34 / 16.32	14.73 / 14.53 / 14.85	11.30 / 11.20 / 11.02	10.04 / 10.57 / 10.57	**7.97 / 8.59 / 8.89**
WCR	36.45 / 32.08 / 30.75	37.57 / 34.42 / 33.40	30.61 / 27.46 / 26.46	30.72 / 27.85 / 27.02	24.26 / 25.04 / 25.33	17.45 / 17.56 / 17.78	13.30 / 15.64 / 16.62	**10.98 / 13.25 / 14.50**
SER	25.41 / 21.66 / 20.79	25.03 / 23.16 / 23.10	20.99 / 18.12 / 17.48	19.80 / 17.48 / 16.72	14.35 / 14.84 / 14.74	12.04 / 11.55 / 11.59	10.82 / 11.03 / 11.18	**8.88 / 9.19 / 9.64**
SECR	44.00 / 38.28 / 36.23	47.46 / 44.36 / 42.52	37.74 / 34.60 / 33.34	38.49 / 35.51 / 34.17	28.96 / 28.05 / 27.52	19.24 / 19.51 / 20.64	14.05 / 15.17 / 16.35	**10.92 / 13.32 / 15.33**
KRCL	24.74 / 20.92 / 19.81	21.57 / 19.11 / 18.68	20.19 / 17.61 / 16.96	18.30 / 16.84 / 16.44	15.51 / **15.86 / 15.76**	18.90 / 19.17 / 19.31	17.41 / 18.72 / 18.72	**13.61** / 15.95 / 16.57

TABLE VI: Zone-wise Root Mean Square Error (RMSE) (in hours) across different delay prediction models

Zones	HA	LSTM	GRU	RF	STGCN	ASTGCN	TSTGCN	RSTGCN (ours)
SCR	0.445 / 0.417 / 0.408	0.424 / 0.401 / 0.391	0.414 / 0.378 / 0.367	0.389 / 0.363 / 0.354	0.456 / 0.449 / 0.450	0.364 / 0.339 / 0.333	0.362 / 0.337 / 0.328	**0.328 / 0.309 / 0.311**
NR	0.714 / 0.666 / 0.645	0.762 / 0.724 / 0.693	0.665 / 0.612 / 0.590	0.634 / 0.585 / 0.563	0.493 / 0.511 / 0.518	0.536 / 0.485 / 0.461	0.538 / 0.488 / 0.463	**0.505 / 0.463 / 0.441**
WR	0.361 / 0.332 / 0.325	0.713 / 0.654 / 0.632	0.525 / 0.488 / 0.475	0.314 / 0.286 / 0.277	0.315 / 0.293 / 0.296	0.232 / 0.223 / 0.209	0.242 / 0.232 / 0.216	**0.219 / 0.215 / 0.205**
ECR	0.827 / 0.751 / 0.730	0.918 / 0.865 / 0.836	0.840 / 0.750 / 0.724	0.712 / 0.642 / 0.623	0.524 / 0.515 / 0.530	0.548 / 0.521 / 0.494	0.552 / 0.528 / 0.503	**0.548 / 0.513 / 0.486**
NWR	0.407 / 0.354 / 0.346	0.391 / 0.342 / 0.334	0.378 / 0.318 / 0.308	0.364 / 0.307 / 0.299	0.256 / 0.253 / 0.253	**0.235 / 0.230 / 0.226**	0.266 / 0.250 / 0.242	0.245 / 0.235 / 0.230
SR	0.547 / 0.521 / 0.515	0.613 / 0.659 / 0.628	0.503 / 0.474 / 0.467	0.453 / 0.431 / 0.427	0.478 / 0.494 / 0.484	0.337 / 0.308 / 0.321	0.376 / 0.343 / 0.352	**0.332 / 0.297 / 0.310**
NFR	0.460 / 0.411 / 0.393	0.440 / 0.396 / 0.378	0.434 / 0.377 / 0.354	0.411 / 0.363 / 0.344	0.477 / 0.414 / 0.386	0.339 / 0.300 / **0.275**	0.358 / 0.310 / 0.285	**0.319 / 0.302** / 0.282
CR	0.717 / 0.668 / 0.644	0.916 / 0.872 / 0.829	0.674 / 0.618 / 0.598	0.636 / 0.587 / 0.568	0.640 / 0.661 / 0.678	0.556 / 0.517 / 0.506	0.512 / 0.490 / 0.487	**0.510 / 0.484 / 0.478**
ECOR	0.468 / 0.428 / 0.416	0.509 / 0.467 / 0.452	0.432 / 0.390 / 0.378	0.402 / 0.368 / 0.359	0.412 / 0.417 / 0.412	0.401 / 0.359 / 0.342	0.379 / 0.343 / 0.330	**0.366 / 0.335 / 0.318**
NCR	0.773 / 0.692 / 0.663	0.734 / 0.668 / 0.643	0.705 / 0.618 / 0.585	0.675 / 0.597 / 0.568	0.522 / 0.521 / 0.527	0.538 / 0.469 / 0.475	0.547 / 0.476 / 0.472	**0.530 / 0.464 / 0.469**
SWR	0.355 / 0.318 / 0.307	0.330 / 0.308 / 0.298	0.331 / 0.290 / 0.277	0.310 / 0.274 / 0.264	0.320 / 0.285 / 0.272	0.249 / 0.234 / 0.229	0.256 / 0.231 / 0.223	**0.237 / 0.220 / 0.216**
ER	0.654 / 0.616 / 0.600	0.853 / 0.814 / 0.812	0.632 / 0.597 / 0.588	0.561 / 0.522 / 0.509	0.411 / 0.389 / 0.406	0.453 / 0.407 / 0.393	0.437 / 0.389 / 0.376	**0.427 / 0.385 / 0.374**
NER	0.919 / 0.828 / 0.822	0.916 / 0.835 / 0.832	0.870 / 0.760 / 0.743	0.853 / 0.746 / 0.732	**0.500 / 0.474 / 0.469**	0.596 / 0.523 / 0.515	0.611 / 0.533 / 0.525	0.559 / 0.497 / 0.497
WCR	0.666 / 0.617 / 0.606	0.640 / 0.614 / 0.606	0.614 / 0.560 / 0.546	0.581 / 0.536 / 0.524	0.609 / 0.613 / 0.611	0.528 / 0.497 / 0.478	0.518 / 0.489 / 0.472	**0.501 / 0.476 / 0.463**
SER	0.670 / 0.605 / 0.581	0.823 / 0.781 / 0.786	0.628 / 0.555 / 0.531	0.597 / 0.528 / 0.504	**0.433 / 0.449** / 0.460	0.544 / 0.478 / 0.466	0.530 / 0.466 / **0.454**	0.502 / 0.448 / 0.449
SECR	0.831 / 0.757 / 0.733	0.811 / 0.756 / 0.748	0.766 / 0.685 / 0.660	0.713 / 0.647 / 0.628	0.698 / 0.688 / 0.695	0.664 / 0.581 / 0.577	0.629 / 0.543 / 0.544	**0.594 / 0.531 / 0.541**
KRCL	0.680 / 0.600 / 0.575	0.607 / 0.548 / 0.531	0.606 / 0.527 / 0.505	0.577 / 0.507 / 0.487	0.561 / 0.588 / 0.570	0.589 / 0.545 / 0.528	0.568 / 0.536 / 0.512	**0.532 / 0.515 / 0.505**

the baselines, benefiting from its inter-station distance modeling, highlighting efficacy of appropriate features.

As depicted from the results, *RSTGCN, our proposed model, consistently achieves the lowest prediction errors across most zones and forecasting horizons*. Its robust performance across all 3 forecast horizons demonstrates strong generalization across varying temporal contexts and zones. This can be attributed to its incorporation of both inter-station distances and train frequency, as well as the newly added features.

Results on the whole IRN: Table VII reports forecasting performance for RSTGCN, TSTGCN and ASTGCN for next 1, 2, and 3 hours on whole IRN. *Avg* denotes metric values averaged over 3 prediction horizons and *(+Impvt.)* denotes RSTGCN's improvement over each model. Here we observe that RSTGCN outperforms the rest baselines with significant mergin for individual horizons as well as in aggregate (with 18-21% improvement in MAE, 14-24% improvement in MAPE and 1-8% improvement on RMSE). It further confirms that the advantages of feature integration and architectural modification extend to large-scale settings (entire IRN) as well.

TABLE VII: Forecasting performance over the entire IR Network for the next 1, 2, and 3 hours with the best-performing 2 baselines. Results are averaged over 3 prediction steps, and (+ Impvt.) denotes the improvement achieved by RSTGCN over each baseline for the corresponding metric.

	MAE	MAPE	RMSE
ASTGCN	0.087 / 0.093 / 0.095	9.71 / 11.26 / 11.62	0.444 / 0.408 / 0.396
Avg + Impvt.	0.092 (+ **18.47%**)	10.86 (+ **14.91%**)	0.416 (+ **8.65%**)
TSTGCN	0.094 / 0.096 / 0.097	11.74 / 12.44 / 12.42	0.393 / 0.382 / 0.379
Avg + Impvt.	0.096 (+ **21.86%**)	12.2 (+ **24.26%**)	0.385 (+ **1.29%**)
RSTGCN	0.069 / 0.078 / 0.079	8.12 / 9.71 / 9.89	0.389 / 0.378 / 0.373
Avg	**0.075**	**9.24**	**0.38**

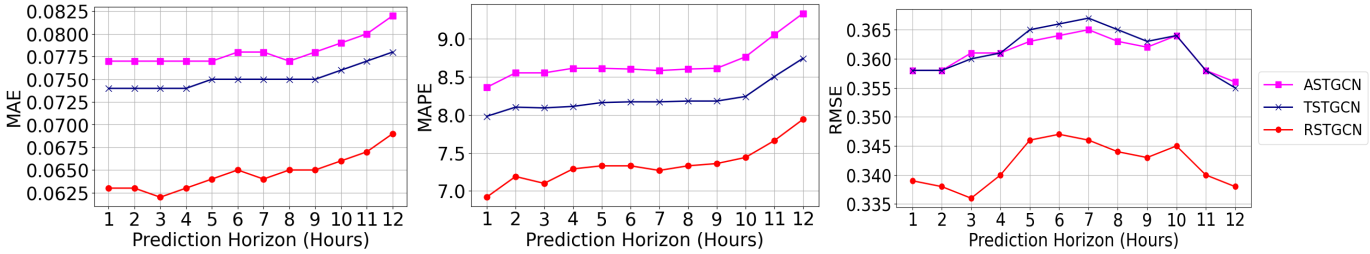


Fig. 4: Delay prediction performance of RSTGCN with two best baselines on the largest zone (SCR) for long-term prediction (up to 12 hours) for all the metrics (averaged over all stations in zone SCR). RSTGCN consistently outperforms baselines.

TABLE VIII: Error analysis: Predictions of our model and best baselines for 1-hour, 2-hour, 3-hour forecasting horizon, shown for some selected stations and timestamps. The predicted delays that are closest to the ground truth are in bold. The top row shows stable or increasing delays where RSTGCN performs well, whereas the bottom row shows irregular fluctuations where other models perform better at times.

Model	[1-hr, 2-hr, 3-hr] delays	[1-hr, 2-hr, 3-hr] delays
	NDLS – NR Zone	**MDU – SR Zone**
Ground Truth	[0.22, 1.43, 1.87]	[0.32, 0.05, 0.10]
ASTGCN	[0.17, 1.18, 0.92]	[0.24, 0.21, 0.29]
TSTGCN	[0.57, 1.68, 0.89]	[0.27, 0.17, 0.19]
RSTGCN	[**0.19, 1.57, 1.62**]	[**0.32, 0.08, 0.12**]
	GHY – NFR Zone	**LJN – NER Zone**
Ground Truth	[0.29, 0.0, 4.11]	[1.15, 2.58, 0.0]
ASTGCN	[0.42, **0.35**, **0.69**]	[1.09, **0.25**, 0.21]
TSTGCN	[0.35, 0.63, 0.47]	[1.05, 0.22, **0.16**]
RSTGCN	[**0.28**, 0.72, 0.02]	[**1.17**, 0.24, 0.26]

F. Long-horizon forecasting

We further evaluate the long-horizon forecasting performance (for 1–12 hours) of RSTGCN and the next best performing baselines (ASTGCN and TSTGCN) on SCR, the largest zone with 499 stations. Figure 4 reports the delay prediction performance in terms of MAE, MAPE, and RMSE, as the prediction horizon varies from 1 to 12 hours. For the i-th hour, the metrics averaged over 1 to i-th hour are reported. We observe that the performance of all models deteriorates (i.e., prediction error increases) with increasing horizon. However, RSTGCN manages to control the degradation better than the other models and maintains superiority with consistent performance gap for all horizons.

G. Error analysis

Till now, we have presented results aggregated over all stations in a zone and the full IRN. Now we check the predictions for some individual stations. Table VIII shows the ground truth mean arrival delays and the predictions of RSTGCN and the two best baselines for 1-hour, 2-hour, 3-hour forecasting horizon, for some selected stations and timestamps.

Simpler cases: The top row shows simpler cases where delay is consistently increasing with the forecasting horizon, or is relatively stable. In such cases, RSTGCN consistently outperform the baselines.

Complex cases: The bottom row contains complex scenarios where the delay follows rather inconsistent or irregular patterns with abrupt/unpredictable fluctuations. In such cases, sometimes TSTGCN or ASTGCN outperforms RSTGCN; however, all models perform poorly in such cases (predicted delays very different from the ground truth).

As the majority of instances (station, timestamp) in IRN follow the first scenario (the highly fluctuating cases being much less frequent), RSTGCN outperforms the baseline models for a majority of the instances in our test set.

H. Ablation Study

We perform an ablation study to investigate the contributions of (1) proposed features and (2) proposed architectural modifications in RSTGCN. The study is conducted on two medium-sized regions: Southern Railway (316 stations) and North Eastern Railway (218 stations).

Feature variants: Table IX presents 3-step prediction results for $\text{RSTGCN}_{\text{AvgDelay}}$ (with 2 features: average arrival and departure delay), $\text{RSTGCN}_{\text{AvgDelay+Headway}}$ (with 3 features: average arrival and departure delay and headway), $\text{RSTGCN}_{\text{AvgDelay+TotDelay}}$ (with 4 features: average and total arrival and departure delay), and $\text{RSTGCN}_{\text{AvgDelay+Headway+TotDelay}}$ (with all 5 features included). We compare these variations with the best baseline TSTGCN. In all the variants of RSTGCN, the architectural modifications described earlier are included. We observe that the addition of average delay improves the model over TSTGCN, whereas the addition of hourly headway and total delay further improves the model performance over $\text{RSTGCN}_{\text{AvgDelay}}$, establishing the contribution of each feature. Addition of total delay yields the maximum improvement among the proposed features. Finally, $\text{RSTGCN}_{\text{AvgDelay+Headway+TotDelay}}$ results in best performance, combining all the proposed features.

Model Variants: Table X presents the impact of individual architectural modifications. We begin by $\text{RSTGCN}_{\text{Base}}$, a version of RSTGCN without any architectural modifications but including all proposed features. $\text{RSTGCN}_{\text{sattn}}$ and $\text{RSTGCN}_{\text{ReLU}}$ respectively combine modified spatial attention and ReLU as activation in the last layer over $\text{RSTGCN}_{\text{Base}}$. We observe that $\text{RSTGCN}_{\text{ReLU}}$ achieves higher performance gain over $\text{RSTGCN}_{\text{sattn}}$. Finally, combining both the updated spatial attention mechanism and ReLU in RSTGCN results in the best performance, highlighting the utility of combining both architectural changes together.

TABLE IX: Ablation study with feature variants upto 1-hour / 2-hour / 3-hour forecasts in comparison with the best baseline. The subscripts in the column headers denote the features used in the corresponding variant of RSTGCN. All the variants of RSTGCN use the updated spatial attention and ReLU function. The best results are shown in **bold**, the second-best are underlined, and the third-best are *italicized*.

Zones	TSTGCN	$\text{RSTGCN}_{\text{AvgDelay}}$	$\text{RSTGCN}_{\text{AvgDelay+Headway}}$	$\text{RSTGCN}_{\text{AvgDelay+TotDelay}}$	$\text{RSTGCN}_{\text{AvgDelay+Headway+TotDelay}}$
			MAE		
SR	0.068 / 0.083 / 0.088	0.061 / 0.072 / 0.075	*0.059 / 0.067 / 0.070*	0.055 / 0.066 / 0.068	**0.054 / 0.063 / 0.067**
NER	0.099 / 0.098 / 0.100	*0.083 / 0.088* / 0.093	0.081 / 0.082 / *0.090*	0.081 / 0.082 / 0.086	**0.078 / 0.079 / 0.084**
			MAPE (%)		
SR	9.57 / 13.72 / 14.50	9.47 / 12.67 / 12.85	*9.04* / 11.12 / 11.66	**8.58** / *11.64 / 11.76*	8.74 / **11.07 / 11.52**
NER	10.04 / 10.57 / 10.57	8.64 / 9.63 / 10.02	8.29 / *9.01 / 9.59*	*8.36* / 8.95 / 9.07	**7.97 / 8.59 / 8.89**
			RMSE		
SR	0.376 / 0.343 / 0.352	0.378 / 0.338 / 0.340	*0.361 / 0.318 / 0.325*	**0.329** / 0.299 / 0.313	0.332 / **0.297 / 0.310**
NER	0.611 / 0.533 / 0.525	0.606 / 0.532 / 0.526	*0.581 / 0.512 / 0.510*	0.575 / 0.508 / 0.505	**0.559 / 0.497 / 0.497**

TABLE X: Ablation study with model variants upto 1-hour / 2-hour / 3-hour forecasts in comparison with the best baseline. The subscripts in the column headers indicate the specific architectural modifications applied in the corresponding variant of RSTGCN. All the variants of RSTGCN use all five features.

Zones	TSTGCN	$\text{RSTGCN}_{\text{Base}}$	$\text{RSTGCN}_{\text{SAttn}}$	$\text{RSTGCN}_{\text{ReLU}}$	$\text{RSTGCN}_{\text{SAtt+ReLU}}$
			MAE		
SR	0.068 / 0.083 / 0.088	0.071 / 0.078 / 0.081	*0.065 / 0.076 / 0.079*	0.060 / 0.070 / 0.072	**0.054 / 0.063 / 0.067**
NER	0.099 / 0.098 / 0.100	0.102 / 0.097 / 0.098	*0.098 / 0.096 / 0.098*	0.079 / 0.083 / 0.085	**0.078 / 0.079 / 0.084**
			MAPE (%)		
SR	9.57 / 13.72 / 14.50	10.52 / 12.87 / 13.24	9.12 / *12.24 / 12.64*	*9.69* / 12.01 / 12.24	**8.74 / 11.07 / 11.52**
NER	10.10 / 10.57 / 10.57	10.32 / 10.45 / 10.36	*10.06 / 10.27 / 10.31*	8.06 / 9.14 / 8.95	**7.97 / 8.59 / 8.89**
			RMSE		
SR	0.374 / 0.343 / 0.352	*0.345 / 0.308* / 0.321	0.351 / 0.312 / *0.320*	0.335 / 0.307 / 0.319	**0.332 / 0.297 / 0.310**
NER	0.611 / 0.533 / 0.525	0.608 / 0.529 / 0.519	*0.605 / 0.527 / 0.518*	0.565 / 0.506 / 0.503	**0.559 / 0.497 / 0.497**

VII. Conclusion

In this work, we address the problem of station-wise mean arrival delay prediction over large-scale railway networks, with a focus on the Indian Railway system. We proposed RSTGCN, a spatio-temporal graph convolutional network that integrates domain-informed features and explicitly incorporates train frequency into spatial dependency modeling to better capture delay propagation dynamics. Extensive experiments on a nationwide Indian Railway delay dataset demonstrate that RSTGCN consistently outperforms state-of-the-art baselines across multiple evaluation settings, achieving improvements of up to 18% in MAE, 14% in MAPE, and 1–8% in RMSE on the IRN. Furthermore, the nationwide railway delay dataset introduced in this work will provide a strong foundation for future research and real-world deployment.

Limitations and future work: The dataset for this paper spans one month only; including exogenous factors such as seasonal weather variations and festive travel patterns are potential future works. Also, our dataset includes express trains only, and does not consider freight and 'local' (short distance) trains. Further, we observe a limited set of irregular and highly unpredictable delay scenarios in which performance degrades for all models, typically caused by sudden disruptions such as system failures or unforeseen events; addressing such extreme/unforeseen cases remains an important direction for future work.

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